Name_____ Date_____ Period_____

LESSON 4-3 · HOMEWORK PACKET · SHEET 1 OF 2

Estimate with Benchmarks

Maryland Standard 6.AT.4 · Learning target – I can estimate a percent by comparing an amount to benchmark percents.

REMEMBER

- **Benchmarks:** 10%, 25%, 50%, 75%, 100%. Build others from 10%: 20% = 2 × 10%, 30% = 3 × 10%.
- Find benchmark amounts of the **whole**. Which one is the part **about** equal to?
- A **double number line** lines up amounts with percents.

WORKED EXAMPLE

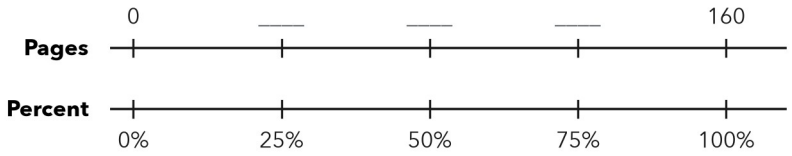
A water jug holds 40 cups. There are 29 cups left. About what percent of the jug is left?

Step 1 Benchmarks of 40: 50% = 20 and 75% = 30.
Step 2 29 is very close to 30.
Step 3 So 29 out of 40 is about 75%.
Answer **about 75%**

PRACTICE

show your work · label units

1. **Reading Goal.** Jada's goal is to read 160 pages. She has read 42 pages. Fill in the double number line. Then estimate the percent of her goal she has read.



About_____ %

2. About what percent? Name the closest benchmark: 10%, 25%, 50%, 75%, or 100%.

a. 7 out of 60 ≈ ___ %
b. 38 out of 50 ≈ ___ %
c. 23 out of 90 ≈ ___ %
d. 19 out of 40 ≈ ___ %
e. 49 out of 50 ≈ ___ %
f. 11 out of 45 ≈ ___ %
g. 16 out of 150 ≈ ___ %
h. 61 out of 80 ≈ ___ %

3. **Class Fundraiser.** A class wants to raise $500. So far they have raised $96. Use 10% of the goal to estimate the percent they have raised.

a. 10% of $500 = _____
b. About what percent have they raised? _____ %

4. **Bus Seats.** A bus has 48 seats. 35 seats are filled. About what percent of the seats are filled? Name the benchmark you used.

About_____ %

5. **Compare.** Which is closer to 50%: 26 out of 50, or 26 out of 60? Explain how you know.

Frame *I know this because ___.*

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Look at a battery icon at home. Estimate its percent and name your benchmark.

Name_____ Date_____ Period_____

LESSON 4-3 · HOMEWORK PACKET · SHEET 2 OF 2

Estimate a Percent of a Number

Maryland Standard 6.AT.4 · Learning target – I can estimate a percent of a number using friendly percents and rounded amounts.

REMEMBER

- Round to **friendly numbers** first.
- 10%: divide by 10. 5% = half of 10%.
- 25% = 1/4, 50% = 1/2, 75% = 3/4.

WORKED EXAMPLE

Estimate 24% of 83.

Step 1 Round: 25% of 80.

Step 2 25% = 1/4, and 80 ÷ 4 = 20.

Answer **about 20**

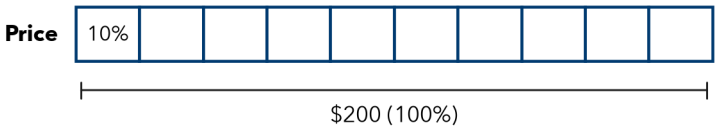
PRACTICE

show your work · label units

1. Estimate. Round to friendly numbers first.

a. 11% of 59 ≈ ___ **b.** 74% of 41 ≈ ___ **c.** 26% of 203 ≈ ___

2. **Sneaker Sale.** Sneakers cost $198 and are 30% off. Round to $200. Each part of the tape is 10% of $200.



a. About how many dollars is one part (10%)? _____

b. About how much is the discount (30%)? _____

3. **School Lunch.** About 48% of 412 students buy lunch. About how many students is that?

About_____ students

4. **Apple Orchard.** A farm picked 798 apples. About 24% of them were too small to sell. About how many apples were too small?

About_____ apples

5. **Find the Error.** Mia estimated 19% of 512 by writing 20 × 500 = 10,000. Explain her mistake. Give a good estimate.

_____

_____

TEST PRACTICE

choose the best answer

6. Which is the best estimate of 26% of 79?

A 2 **B** 20 **C** 26 **D** 40

7. A coat costs $119. It is 40% off. Which is the best estimate of the discount?

A $4.80 **B** $40 **C** $48 **D** $72

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Find a sale sign. Estimate the discount out loud with someone at home.